

Sister Nivedita University

Undergraduate course structure for Biotechnology

As per NEP 2020 regulation and according to UGC-CBCS



Course structure for

B.Sc. in Biotechnology

And

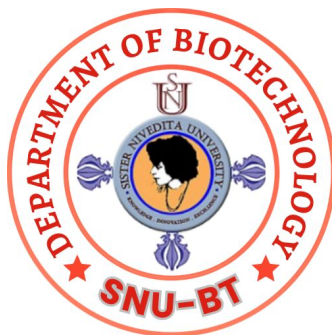
B.Sc. Honours in Biotechnology / B.Sc. Honours with Research in Biotechnology

This curriculum is duly approved by the Board of Studies, Department of Biotechnology

Sister Nivedita University

Undergraduate course structure for Biotechnology

As per NEP 2020 regulation and according to UGC-CBCS



Course structure for

B.Sc. in Biotechnology

And

B.Sc. Honours in Biotechnology / B.Sc. Honours with Research in Biotechnology

Category definition with credit breakup

Semester	Minor Related Courses			Major Related Courses							University Course		Credit / Semester
	MDC	MN	VOC		MC/ME	Project/ Course	SEC	SIP	AEC		VAC		
			MN Related	MC/ME Related					MS	BC/EC/FL			
I					12 (4+4+4)		3 (1+1+1)				3 (2+1)	18	
II	3	4	2		8 (4+4)		2 (1+1)				3 (2+1)	22	
III	3	4	2		8 (4+4)		2 (1+1)		1			20	
IV	3	4	2		8 (4+4)		2 (1+1)		1			20	
V				6 (2+2+2)	12 (4+4+4)				1	1		20	
VI					12 (4+4+4)			4	1	3 (2 + 1)		20	
3-Year Degree	9	24			60		9	4	8		6	120	
VII		4 + 4			12 (4+4+4)							20	
VIII					8 (4+4+4)	12						20	
4-Year Degree	9	32			80	12	9	4	8		6	160	



Abbreviation	Full Form	Abbreviation	Full Form
MC	Major Compulsory	VOC	Vocational Course
ME	Major Elective	MDC	Multidisciplinary Course
SEC	Skill Enhancement Course	MS	Mentored Seminar
SIP	Summer Internship	BC	Business Communication
AEC	Ability Enhancement Course	CE	Communicative English
VAC	Value Added Course	EVS	Environmental Science
MN	Minor	IKS	Indian Knowledge System

Objectives and outcomes of the program and every courses

VISION

- To develop competent and ethically responsible Biotechnologists with advanced scientific and technological expertise, capable of contributing to health, agriculture, environmental sustainability, and industrial innovation.
- To emerge as a centre of excellence in Biotechnology education and research in the Eastern India region, advancing biotechnological innovation and addressing unmet medical needs on both local and global platforms.
- To construct transformative and impactful knowledge that addresses the needs of diverse communities while promoting inclusion, diversity, and the advancement of an enlightened and equitable society.

MISSION

- To offer various undergraduate, postgraduate and PhD programs that foster innovative research, critical thinking, effective communication, and industry engagement, enabling students to acquire relevant technical, interpersonal, and managerial skills for professional and academic excellence.
- To foster a diverse and interdisciplinary research environment by leveraging faculty expertise across computational drug design, Plant Biotechnology, Nanomedicine, and Cancer research, thereby advancing innovative research in Health and Translational Science.
- To empower learners to become technically proficient, emotionally intelligent, socially responsible, and ethically grounded individuals who are future-ready, entrepreneurial, globally minded, and committed to fostering a sustainable and inclusive society.

PROGRAMME EDUCATIONAL OBJECTIVES:

PEO1: To prepare graduates with strong foundational knowledge in biological sciences and biotechnology, capable of demonstrating scientific reasoning, laboratory skills, and ethical values to address challenges in health, agriculture, industry, and the environment.

PEO2: To develop graduates who pursue higher education, research, or entry-level professional careers with competence in communication, teamwork, critical thinking, and social responsibility.

PROGRAM SPECIFIC OUTCOME (PSOs)

PSO1: Demonstrate core concepts of molecular biology, microbiology, cell biology, and genetics to describe biological systems and solve basic biotechnological problems.

PSO2: Demonstrate proficiency in standard laboratory techniques, data interpretation, and the use of biotechnological tools required for research, diagnostics, and industry operations.

PROGRAMME OUTCOMES:

PO1: Ability to design, implement, and appraise basic biotechnological experiments and projects that address societal, healthcare, agricultural, and environmental needs.

PO2: Ability to use modern laboratory tools, analytical instruments, and computational techniques relevant to biotechnology.

PO3: Ability to assess the impact of contemporary biotechnological developments on society, healthcare, industry, and ethical practice.

PO4: Ability to recognize the sustainability, biosafety, and environmental implications of biotechnological solutions.

PO5: Ability to follow prescribed laboratory norms, biosafety guidelines, ethical principles, and regulatory frameworks.

PO6: Ability to work effectively as an individual and as a member of a multidisciplinary team.

PO7: Ability to communicate scientific ideas and experimental outcomes clearly through oral, written, and visual presentations to diverse audiences.

PO8: Ability to recognize the need for lifelong learning and to update skills in pace with advancing biotechnological research and industry practices.

Credit Definition

Type	Duration (in hours)	Credit
Lecture (L)	1	1
Tutorial (T)	1	1
Practical (P)	2	1

Category	Course name	Credit	Teaching Scheme		
			L	T	P
Semester I					
MC 1	BIOCHEMISTRY	4	4	0	0
MC 2	BASICS OF MICROBIOLOGY	4	4	0	0
MC3	INTRODUCTION TO BIOTECHNOLOGY AND CELLULAR SYSTEM	4	4	0	0
SEC1	BIOCHEMISTRY LAB	1	0	0	2
SEC2	BASICS OF MICROBIOLOGY LAB	1	0	0	2
SEC3	CELL BIOLOGY LAB	1	0	0	2
VAC1	VOCATIONAL COURSE-EXTRA ACADEMIC ACTIVITY	1	1	0	0
VAC2	ENVIRONMENTAL SCIENCE-I	2	2	0	0
Total Credit = 18			Teaching Hour = 21		

Semester –I
MC-1: Biochemistry
Credit: 4 (4L-0T-0P)

Course Component: Theory

Lecture Hour: 48

Course Outcomes:

After successfully completing the course the student will be able to

CO 1 To explain the origin, organization, and evolution of life, and the role of water in biological systems.

CO 2 To describe the molecular forces and interactions responsible for the structure and stability of biomolecules.

CO 3 To illustrate the structure, properties, and biological significance of carbohydrates, amino acids, and proteins.

CO 4 To analyze the classification, structure, and chemical properties of lipids and their biological functions.

CO 5 To explain the structure, properties, and functions of nucleic acids and their role in genetic information storage and transmission.

MAPPING OF COs WITH POs AND PSOs

COURSE OUTCOMES	PROGRAMME OUTCOMES								PROGRAMME SPECIFIC OUTCOMES	
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2
CO1	1	1	1	1	1	1	1	1	2	2
CO2	1	1	1	1	1	1	1	2	2	2
CO3	2	2	2	1	1	2	2	2	3	3
CO4	3	3	3	1	1	2	2	2	3	3
CO5	3	3	3	1	1	2	2	2	3	3

1. LOW 2. MODERATE 3. SUBSTANTIAL

Teaching Topics

Unit 1: Organization in Nature:

[4 L]

The Scientific Method; Linnaean hierarchical classification system; Evolution as a Unifying Theme of Life; Origins of Life: Early Chemical and Biological Evolution; Oparin-Haldane model; Miller-Urey experiment; Structure of Water; Properties of water

Unit 2: Forces in Biomolecules:

[4 L]

Types of interaction between atoms, van der Waals interactions, Coulombic, dipole-dipole, hydrophobic interaction, hydrogen bond.

Unit 3: Carbohydrates:

[10 L]

Definition, classification and structural concept of: Monosaccharides: Hexoses (only Glucose), Pentoses (Ribose, Ribulose, Xylose) Disaccharides: Sucrose, Lactose, Maltose Amino Sugars: Glucosamine, Muramic Acid Inversion (hydrolysis) of cane sugar. Chemical reactions of monosaccharides (glucose & fructose i.e., aldose and ketose) with HN03, Br2 - water, HI04,

Phenylhydrazine. Principle of chemical estimation of sugar. Anomeric effect (Methylation effect). Polysaccharides: Chemical structure of Starch (α -amylose, amylopectin), glycogen & cellulose. Smith degradation and enzymic hydrolysis of α -amylose & amylopectin.

Unit 4: Amino Acids:

[8 L]

Definition, classification, structure, and stereochemistry of amino acids; physicochemical properties (ionisation and Biuret reaction) of amino acids. Amphoteric molecule, Zwitterion, pK values; Isoelectric point, Electrophoresis. Separation of amino acids by ion-exchange, Gel filtration, Paper Chromatography, and Thin-Layer Chromatography. Peptides: peptide bond, biologically important peptides (glutathione, oxytocin-important functions).

Unit 5: Proteins:

[8 L]

Classification (Primary, Secondary, Tertiary, Quaternary- definition, examples). Ramchandran plot. Types of proteins: i) Fibrous (α -helix, β - sheet, e.g. collagen): definition and structure. ii) Globular: definition & examples. iii) Simple proteins and conjugated protein: definition & examples—physical denaturation and renaturation, Protein folding.

Unit 6: Lipids:

[6 L]

Definition, nomenclature, classification - (simple, complex, derived lipids - structure & example) phospholipids, glycolipids, - (structure, composition); Soap, surfactants, fatty acid. hydrolysis, saponification, saponification number, I2 number, acetylation, acetyl number, volatile fatty acid number - definition and related problems, Isomerism - cis-trans isomerism. Fatty acids: Saturated (palmitic acid, stearic acid), unsaturated (oleic acid): Structure of free fatty acids. General chemical reaction of fatty acids - esterification. Hydrogenation and halogenations

Unit 7: Nucleic acid:

[8L]

Purine, pyrimidine - definition and structure. Nucleoside, nucleotide: definition and structure. DNA & RNA: Double helical structure. A-DNA, B-DNA & Z-DNA (structure and differences). Chemical Properties: Hydrolysis (acid, alkali), enzymatic hydrolysis of DNA. General structure and types of RNA (tRNA, mRNA, rRNA). Hyperchromic effect. Cot curve, T_m .

Suggested Books:

1. Principles of Biochemistry., A Lehninger.
2. Textbook of Biochemistry., MD Rafi
3. Biochemistry., L Stryer.
4. Harper's Biochemistry., R. K. Murray
5. Biochemistry., Voet and Voet

SEC – 1: Biochemistry Practical Credit: 1 (0L-0T-2P)

Course Outcomes:

After successfully completing the course the student will be able to

CO1: Classify different laboratory instruments (micropipette, balance, pH meter) and explain their principles and safe operating procedures.

CO2: Demonstrate the ability to prepare laboratory buffers, measure pH accurately, and use

basic biochemical tools.

CO3: Examine biomolecules through qualitative tests and differentiate amino acids, reducing sugars, and non-reducing sugars using standard biochemical assays.

CO4: Appraise the accuracy, reliability, and limitations of biochemical test results and appraise procedural deviations during experiments.

CO5: Design a systematic experimental workflow combining multiple biochemical assays and produce properly recorded, interpretable laboratory data.

MAPPING OF COs WITH POs AND PSOs

COURSE OUTCOMES	PROGRAMME OUTCOMES								PROGRAMME SPECIFIC OUTCOMES	
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2
CO1	2	3	1	1	3	1	1	1	2	3
CO2	3	3	1	1	3	1	1	1	2	3
CO3	3	3	1	1	2	1	2	1	3	3
CO4	2	2	2	1	3	1	2	1	2	3
CO5	3	3	1	1	2	2	3	1	3	3

1. LOW

2. MODERATE

3. SUBSTANTIAL

Practicals:

1. Preparation of buffers. Concepts of molarity.
2. pH measurements.
3. pH Titration of the Glycine HCl solution.
4. RBC count using hemocytometer.
5. DNA absorption spectra.
6. Gram staining of bacteria.
7. Estimation of blood glucose.
8. Assay of salivary amylase.
9. Isolation of lecithin, identification by TLC, and its estimation.
10. Estimation of protein and sugar.

MC 2: Basics of Microbiology Credit: 4 (4L-0T-4P)

Course Component: Theory

Lecture Hour: 48

Course Outcomes:

After successfully completing the course the student will be able to

CO1: Recall and explain the fundamental characteristics, structural organization, and classification of microorganisms including bacteria, fungi, viruses, and protozoa.

CO2: Demonstrate basic microbiological concepts to interpret microbial growth, nutrition, reproduction, and interactions with the environment.

CO3: Distinguish between different microbial groups based on morphological, physiological, and biochemical properties.

CO4: Examine simple experimental data or culture results to infer microbial identity, growth patterns, or metabolic activities.

CO5: Propose basic microbiological procedures such as staining, culturing, or sterilization techniques for studying microorganisms safely and effectively.

MAPPING OF COs WITH POs AND PSOs

COURSE OUTCOMES	PROGRAMME OUTCOMES								PROGRAMME SPECIFIC OUTCOMES	
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2
CO1	3	2	2	2	2	–	1	2	3	2
CO2	3	3	2	2	2	–	–	2	3	3
CO3	3	3	2	2	2	–	–	1	3	2
CO4	3	2	2	2	2	1	–	1	3	3
CO5	3	3	2	3	3	1	–	2	3	3

1. LOW 2. MODERATE 3. SUBSTANTIAL

Teaching Topics

Unit 1: Introduction to Microbiology [6L]

History of Microbiology, The Conflict over Spontaneous Generation, Koch's Postulates.

Unit 2: Microbial Cell structure and organization [12L]

Bacteria: Classification, Types, Morphology (Size, Shape and Arrangement) Cell ultra-structure: Peptidoglycan structure and synthesis. Cytoplasmic matrix and components: Inclusion bodies. Reproduction and Growth (Binary Fission, Conjugation and Endospore formation) . Eukaryotic microorganisms: Basic structure and function of algae, fungi and protozoa. Disease caused by them

Unit 3: Microscopy and Staining [10L]

Microscope-Principles, types and applications.. , Stains and Staining Solutions, Definition of Dye and Chromogen. Structure of Dye and Chromophore. Functions of Mordant and Fixative. Natural and Synthetic Dyes. Simple Staining, Differential Staining and Acid Fast Staining with specific examples

Unit 4: Microbial Cultivation and Enumeration [10L]

Cultivation of Microorganisms: Nutritional Requirements and Growth Factors. Classification of Organisms Based on Nutritional Types, Design and Types of Culture Media: Simple, Differential, Selective and Enriched. Concept of Isolation and Methods of Isolation. Pure Culture Techniques. Growth and Enumeration Growth Phases, Growth Curve. Arithmetic Growth and Growth Yield. Measurement of Growth. Chemostat and Turbidostat Enumeration of Microorganisms-Direct and Indirect Methods. Preservation of Cultures-Principle and Methods.

Unit 5: Sterilization Techniques [10L]

Definition: Sterilization and Disinfection. Types and Applications Dry Heat, Steam under pressure, Gases, Radiation and Filtration Chemical Agents and their Mode of Action-

Aldehydes, Halogens, Quaternary Ammonium Compounds, Phenol and Phenolic Compounds, Heavy Metals, Alcohol, Dyes, and Detergents Ideal Disinfectant. Examples of Disinfectants and Evaluation of Disinfectant.

SEC – 2: Basics of Microbiology Practical
Credit: 1 (0L-0T-2P)

Component: Practical

Course Outcomes:

After successfully completing the course the student will be able to

CO1: Identify various sterilization techniques and interpret their role in maintaining aseptic laboratory practices.

CO2: Execute pure culture methods—streak plate, pour plate, stab, and slant—and utilize them to obtain isolated microbial colonies.

CO3: Differentiate microbial populations based on colony characteristics and examine patterns obtained through isolation procedures.

CO4: Assess staining outcomes (Gram, endospore, simple, negative) and judge their relevance in preliminary microbial identification.

CO5: Construct a systematic characterization scheme integrating isolation, staining, and biochemical tests to generate a microbial identity profile.

MAPPING OF COs WITH POs AND PSOs

COURSE OUTCOMES	PROGRAMME OUTCOMES								PROGRAMME SPECIFIC OUTCOMES	
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2
CO1	3	2	1	2	3	1	1	1	3	3
CO2	3	3	1	2	3	2	1	1	3	3
CO3	3	3	1	2	2	1	2	1	3	3
CO4	3	2	2	2	3	1	2	1	3	3
CO5	3	3	2	3	2	2	3	1	3	3

1. LOW

2. MODERATE

3. SUBSTANTIAL

Practicals:

1. Basic sterilization techniques
2. Pure culture techniques; Types of culture method Streak plate, Pour plate, Stab & Slant preparation
3. Isolation of bacteria
4. Gram's staining, endospore, simple, negative.
5. Biochemical characterization of microbes

Suggested Books:

1. Willey, J.M., Sherwood, L.M., & Woolverton, C.J. (2020). Prescott's Microbiology (11th ed.). McGraw-Hill Education.
2. Pelczar, M.J., Chan, E.C.S., & Krieg, N.R. (1993). Microbiology (5th ed.). McGraw-Hill Education.
3. Tortora, G.J., Funke, B.R., & Case, C.L. (2018). Microbiology: An Introduction (12th ed.). Pearson.

4. Madigan, M.T., Bender, K.S., Buckley, D.H., Sattley, W.M., & Stahl, D.A. (2018). Brock Biology of Microorganisms (15th ed.). Pearson.

MC 3: Introduction to Biotechnology & Cellular Systems

Credit: 4 (4L-0T-4P)

Course Component: Theory

Lecture Hour: 48

Course Outcomes:

After successfully completing the course the student will be able to

CO1: Explain the fundamental concepts, historical development, tools, techniques, and scope of biotechnology, including the central dogma and computational biotechnology.

CO2: Describe the major applications of biotechnology in agriculture, medicine, industry, and environmental management, along with concepts of bioethics, biosafety, IPR, and metabolic engineering.

CO3: Illustrate the structure, organization, and diversity of prokaryotic and eukaryotic cells, including cellular components and organelles.

CO4: Explain cellular functions, membrane transport mechanisms, cell communication, energy transformation, and different modes of cell division and differentiation.

CO5: Describe the characteristics, structure, replication, and significance of viruses, viroids, and prions in biological systems.

MAPPING OF COs WITH POs AND PSOs

COURSE OUTCOMES	PROGRAMME OUTCOMES								PROGRAMME SPECIFIC OUTCOMES	
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2
CO1	3	2	1	1	–	1	1	1	3	3
CO2	3	2	3	3	1	1	1	2	3	3
CO3	3	3	1	1	–	1	1	–	3	3
CO4	3	3	2	2	–	2	1	1	3	3
CO5	2	2	2	1	–	1	1	1	3	3

1. LOW 2. MODERATE 3. SUBSTANTIAL

Unit I: Introduction to Biotechnology

[8L]

Definition, scope, and importance of biotechnology, Contributions of major scientists in biotechnology, Central Dogma, Tools and techniques in Biotechnology, Introduction to Computational biotechnology (introductory concept)

Unit 2: Applications of Biotechnology

[12L]

Agricultural biotechnology, Medical biotechnology and vaccines, Industrial biotechnology and fermentation, Environmental biotechnology and bioremediation, Metabolic Engineering, Bioethics and biosafety. Concept of IPR.

Unit 3: Cellular Organization in Living Systems [18L]

Cell theory and levels of biological organization, Comparative account of prokaryotic and eukaryotic cells, Cell size and complexity, Cell membrane and cell wall, Cytoplasm, nucleus and nucleoid, Ribosomes and membrane-bound organelles, Cell appendages: flagella, cilia, and pili, Endosymbiotic theory. Discovery of viruses, General characteristics, Structure of viruses, Bacteriophages, Animal and plant viruses, Viral replication (lytic and lysogenic stages), cytopathic effect, Viroid and prions, Importance of viruses

Unit 4: Cellular Functions [10L]

Membrane transport: diffusion, osmosis, facilitated diffusion, and active transport, Cellular activities and energy transformation, Cellular communication (basic concepts), Binary fission, Mitosis and meiosis, Cell differentiation and cell death mechanism,

Suggested books:

1. **Introduction to Biotechnology** – William J. Thieman & Michael A. Palladino, **3rd Edition**, Pearson Education, 2013.
2. **Molecular Biology of the Cell** – Bruce Alberts et al., **6th Edition**, Garland Science (Taylor & Francis), 2014.
3. **A Textbook of Biotechnology** – R. C. Dubey, **5th Edition**, S. Chand Publishing, 2014.

SEC – 3: Cell Biology Practical Credit: 1 (0L-0T-2P)

Component: Practical

Course Outcomes:

After successfully completing the course the student will be able to

CO1 Students will learn to identify different stages of mitosis through observation of onion root tip squash and permanent slides and relate cell division to growth and reproduction.

CO2 Students will learn to perform cell counting using a haemocytometer and estimate cell density in biological samples.

CO3 Students will learn to follow laboratory safety rules, handle laboratory equipment properly, and apply sterilization techniques for safe laboratory practices.

CO4 Students will learn to understand the basic principles and applications of plant and animal cell culture through laboratory demonstrations.

CO5 Students will learn to prepare buffers and perform agarose gel preparation for basic molecular biology experiments.

MAPPING OF COs WITH POs AND PSOs

COURSE OUTCOMES	PROGRAMME OUTCOMES								PROGRAMME SPECIFIC OUTCOMES	
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2
CO1	3	3	1	1	3	2	1	–	3	3
CO2	2	1	3	2	–	3	1	–	3	3
CO3	2	1	2	3	1	3	1	–	3	3

CO4	3	2	1	2	–	2	1	1	3	3
CO5	2	1	2	2	–	3	1	1	3	3

1. LOW 2. MODERATE 3. SUBSTANTIAL

Practicals:

1. Laboratory safety rules, Handling of equipments and Sterilization techniques
2. Agarose gel and preparations of buffer
3. Cell counting using haemocytometer
4. Demonstration of plant and animal cell culture
5. Observation of mitosis in onion root tip squash / permanent slides

Suggested books:

1. Molecular Biology of the Cell – Bruce Alberts, Alexander Johnson, Julian Lewis, David Morgan, Martin Raff, Keith Roberts, and Peter Walter, 7th Edition, W.W. Norton & Company, 2022.
2. Culture of Animal Cells: A Manual of Basic Technique and Specialized Applications – R. Ian Freshney, 8th Edition, Wiley-Blackwell, 2021.
3. Introduction to Biotechnology – William J. Thieman and Michael A. Palladino, 4th Edition, Pearson Education, 2019.

VAC 2: Environmental Science I

Credit: 2 (2L-0T-0P)

Course Component: Theory

Lecture Hour: 24

Course Outcomes:

After successfully completing the course the student will be able to

CO1: Categorize different spheres of environment and their components.

CO2: Differentiate among perpetual, renewable and non-renewable resources and clarify the approach towards their preservation.

CO3: Recognize the components of atmospheric layers, their specific roles on environment and related problems.

CO4: Appraise different resources viz. water, forest, food, land, mineral and energy and their functions for sustaining life on earth.

CO5: Illustrate the ways of destruction of specific resources and its prevention in turn.

MAPPING OF COs WITH POs AND PSOs

COURSE OUTCOMES	PROGRAMME OUTCOMES								PROGRAMME SPECIFIC OUTCOMES	
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2
CO1	1	0	1	2	0	0	1	2	1	0
CO2	1	0	2	3	0	0	1	2	1	0
CO3	1	0	2	3	0	0	1	1	1	0

CO4	1	0	2	3	0	0	1	2	1	0
CO5	1	0	3	3	1	0	1	2	1	0
1. LOW			2. MODERATE			3. SUBSTANTIAL				

Teaching Topics

Unit-1: Components of Environment and Atmosphere (4L)

Classification of natural resources (four atmospheric spheres); Perpetual, renewable and nonrenewable resources. Names and compositions of different layers of atmosphere; introduction of pollutant and contaminant.

Unit-2: Water resource (4L)

Classification and quantization of water resource; Sweet and saline water; Fresh water resource: surface water, water table and aquifer; use and over use of water; causes of flood and draught; conflict over water; benefit and harms of dams; water harvesting and our responsibilities.

Unit-3: Forest resource (4L)

Classification and importance of forest; use and over exploitation of forests; deforestation: cause (timber extraction, mining and dam construction) and effect (global warming, erosion, lower rain fall); afforestation: forest growth, preservation and laws, tribal people and forest.

Unit-4: Food resource (2L)

Different types of foods their origin and their importance; global food problems; food production and food loss; modern agriculture, GM food.

Unit-5: Land resource (4L)

Importance of land as a resource and its classification; land fertility and degradation; over grazing and over cultivation; effect of pesticide and fertilizer, land erosion, landslide, desertification (by human activity).

Unit-6: Mineral resource (3L)

Introduction of minerals and their classification; use and importance; effect of extracting mineral (mining) on environment.

Unit-7: Energy resource (3L)

Introduction and origin; Perpetual, renewable and non-renewable energy resources; growing energy needs; energy harvesting, green energy, alternative energy resources.

Recommended Book:

1. Textbook for Environmental Studies for Undergraduate Course; Erach Bharucha; UGC, New Delhi and Bharati Vidyapeeth Institute of Environment Education and Research, Pune.